



EMPOWER the Trading World

Infinity Algo Engine\$ Trader Manual

1. Overview

Dear Trader,

When you receive this document, it means you now have access to **Infinity Algo Engine\$** – important step toward becoming a professional trader by **automating your trading methods and signals**. This process helps remove emotional influence, standardize trading behavior, reduce execution errors, and improve consistency. It also allows you to test and optimize your trading approach. Once your trading decision and rules are defined, the system handles the remaining execution according to the configuration you have set.



2. Execution framework

Infinity Algo Engine\$ generates signals based on the analysis of any indicator you choose, calculates entry prices, submits orders according to the configured parameters, and manages the trade state.

The trading process can be divided into 4 layers:

- **Indicator Layer:** The system or analytical method used to identify signals.
- **Decision Layer:** The decision of whether to participate in a trade.
- **Execution Layer:** How orders are placed and executed in the market.
- **Risk Governance Layer:** Controls profit/loss management and trading frequency.

Infinity Algo Engine\$ focuses primarily on the Execution Layer while also supporting the Risk Governance Layer.

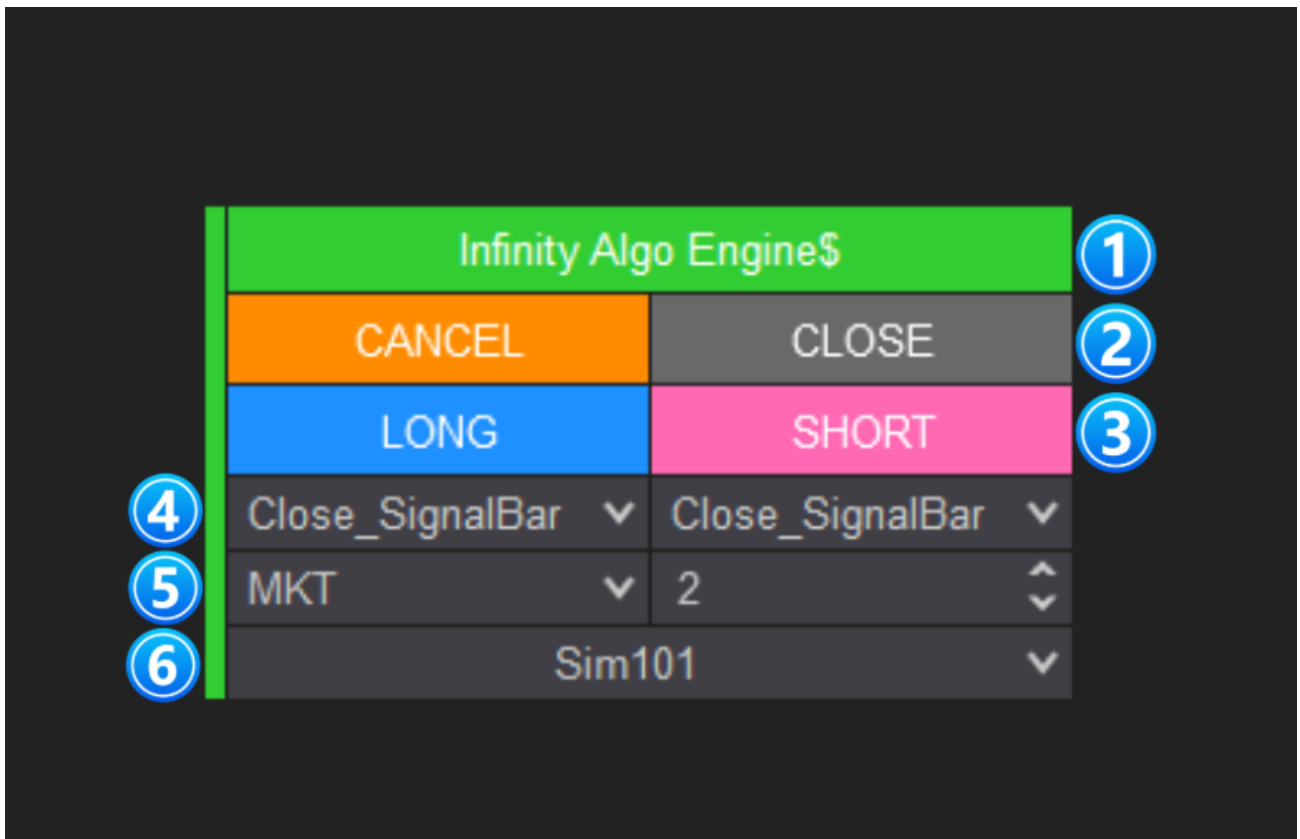
The objective of the system is to ensure that once a trading decision has been made, the order is executed according to the defined rules and configuration, minimizing deviations caused by emotions or manual interaction.

2.1 Interface

Infinity Algo Engine\$ is designed with a simplified interface that focuses on the key functional groups involved in trade execution.

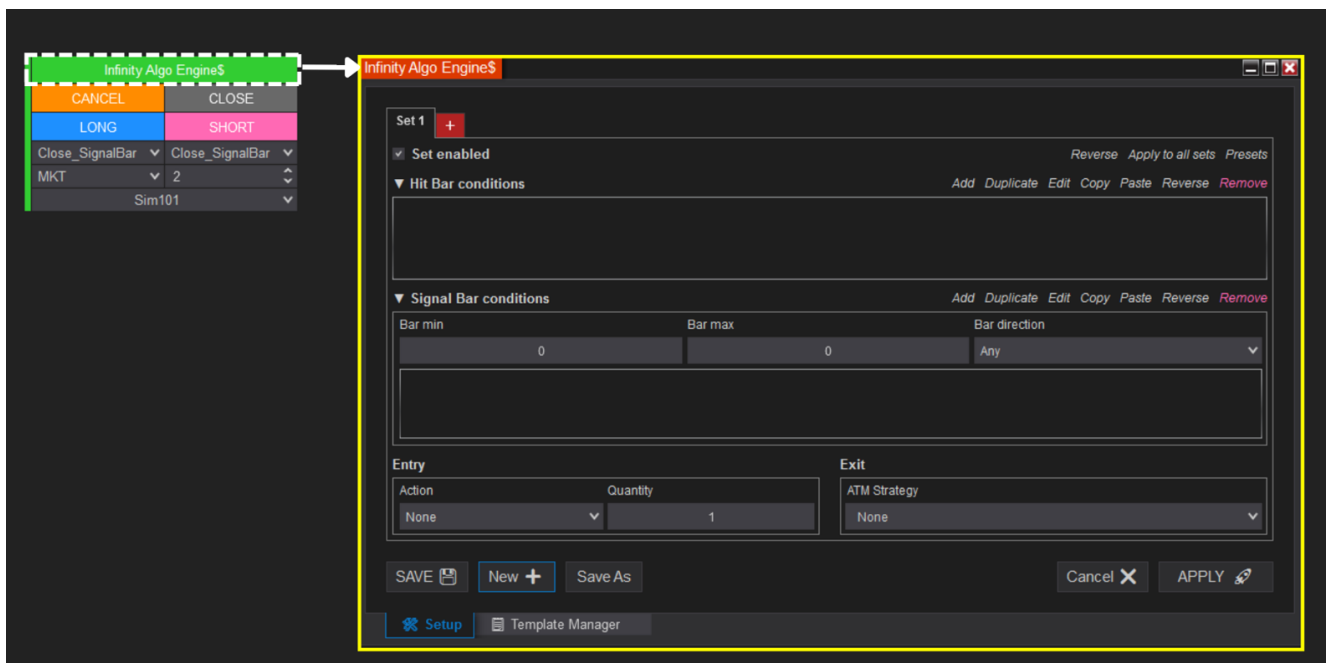
2.1.1 Control Panel

The control panel includes the following sections:



- **Row 1:**

- **"Infinity Algo Engine\$" Button:** Opens the configuration window where you define and manage the system's signal conditions.



- **Row 2:**

- **"CANCEL" Button:** Cancels all active pending orders.
- **"CLOSE" Button:** Closes all open positions and cancels all active pending orders.

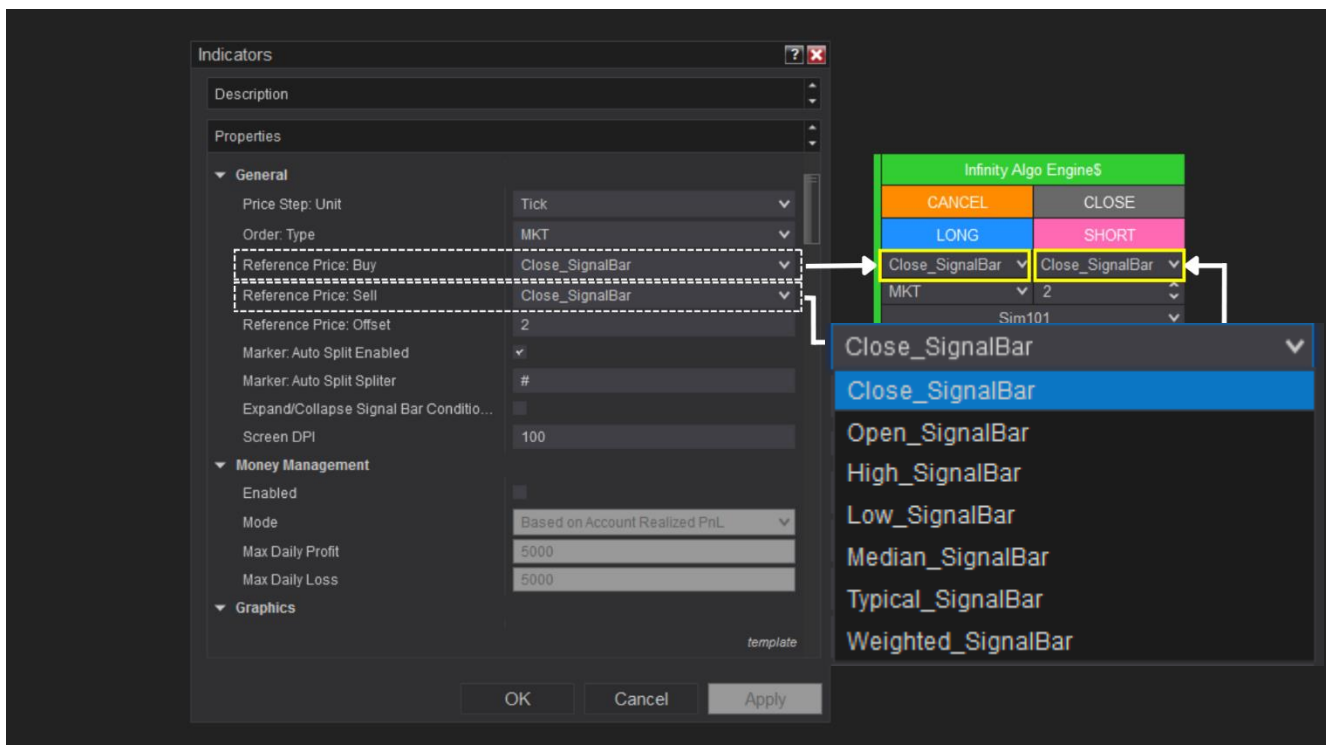
- **Row 3:**

- **"LONG" Button:** When enabled, the system automatically executes Buy orders when the configured Buy conditions are met.
- **"SHORT" Button:** When enabled, the system automatically executes Sell orders when the configured Sell conditions are met.

- **Row 4:**

"Reference Price": The base price from which the offset is applied.

- "Reference Price: Buy": Defines the reference price used for Buy orders.
- "Reference Price: Sell": Defines the reference price used for Sell orders.

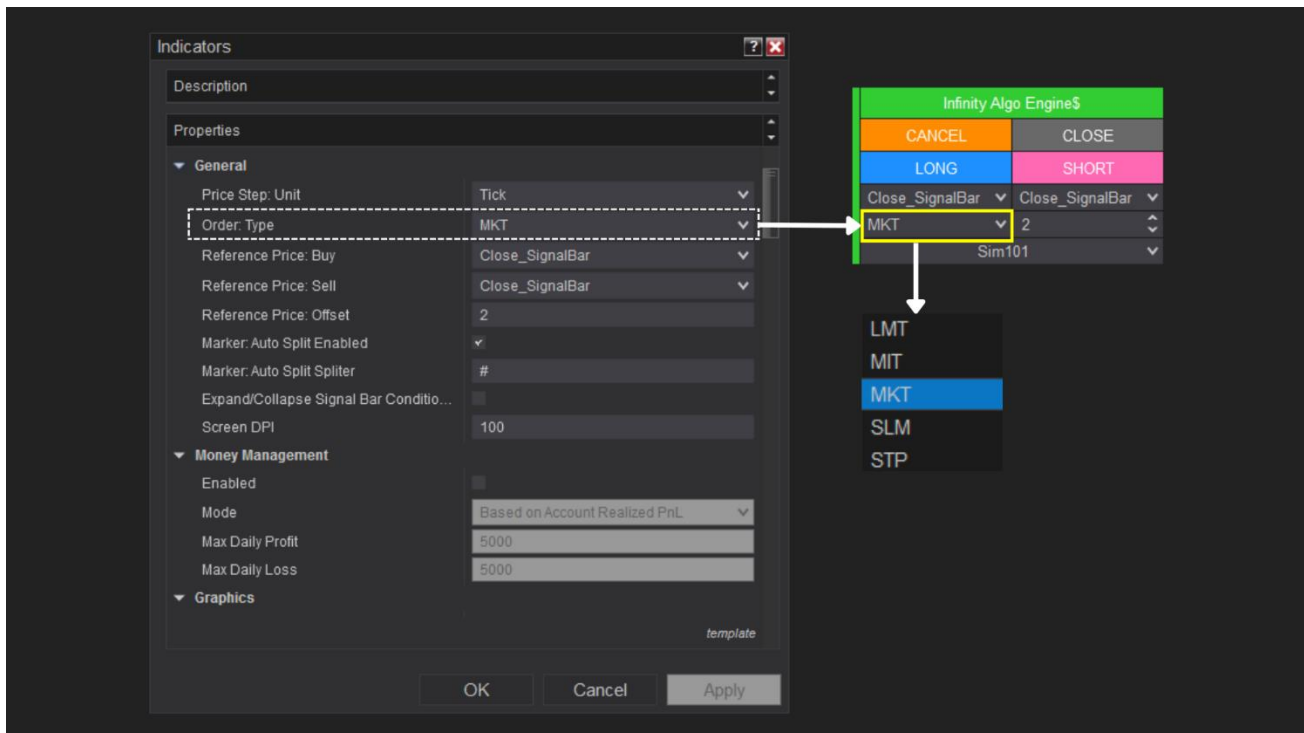


Both Buy and Sell configurations can use one of the following 7 reference price types:

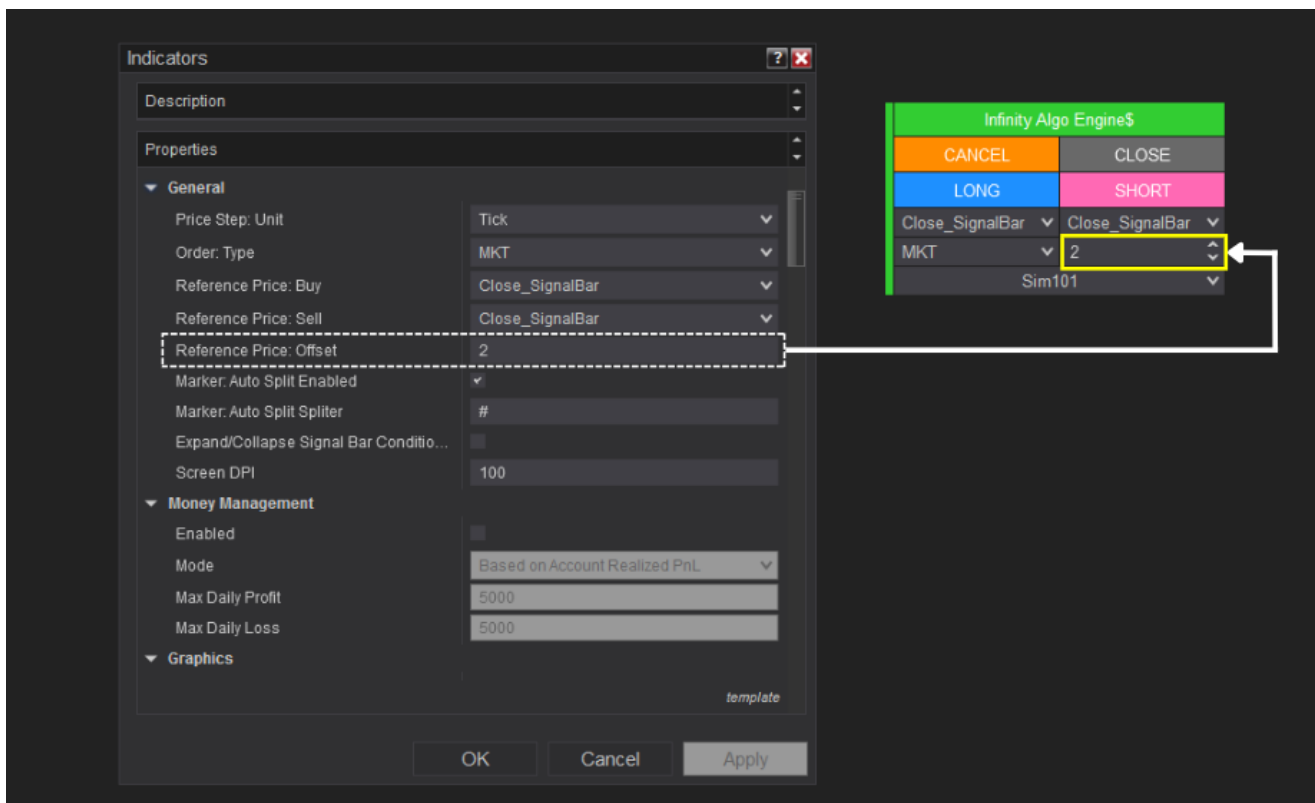
- "Close_SignalBar": Uses the closing price of the signal bar as the reference price.
- "Open_SignalBar": Uses the opening price of the signal bar as the reference price.
- "High_SignalBar": Uses the highest price of the signal bar as the reference price.
- "Low_SignalBar": Uses the lowest price of the signal bar as the reference price.
- "Median_SignalBar": Uses the average of the High and Low prices of the signal bar as the reference price.

- "Typical_SignalBar": Uses the average of the High, Low, and Close prices of the signal bar as the reference price.
- "Weighted_SignalBar": Uses the weighted average of the Close, Low, and High prices of the signal bar as the reference price.

- **Row 5:**



- "Order: Type": Select among the available order types:
 - MKT: Market order.
 - STP: Pending Stop order.
 - SLM: Pending Stop-Limit order.
 - LMT: Pending Limit order.
 - MIT: Pending Market-If-Touched order.



- "Reference Price: Offset Value": The offset (from the reference price) for the **Infinity Algo Engine\$** system to calculate the actual order price. The offset is measured in Tick.

Example: Consider a Buy order with offset = 2 ticks.

- Order type = LMT/MIT: actual order price is 2 ticks BELOW the reference price.
- Order type = SLM/STP: actual order price is 2 ticks ABOVE the reference price.

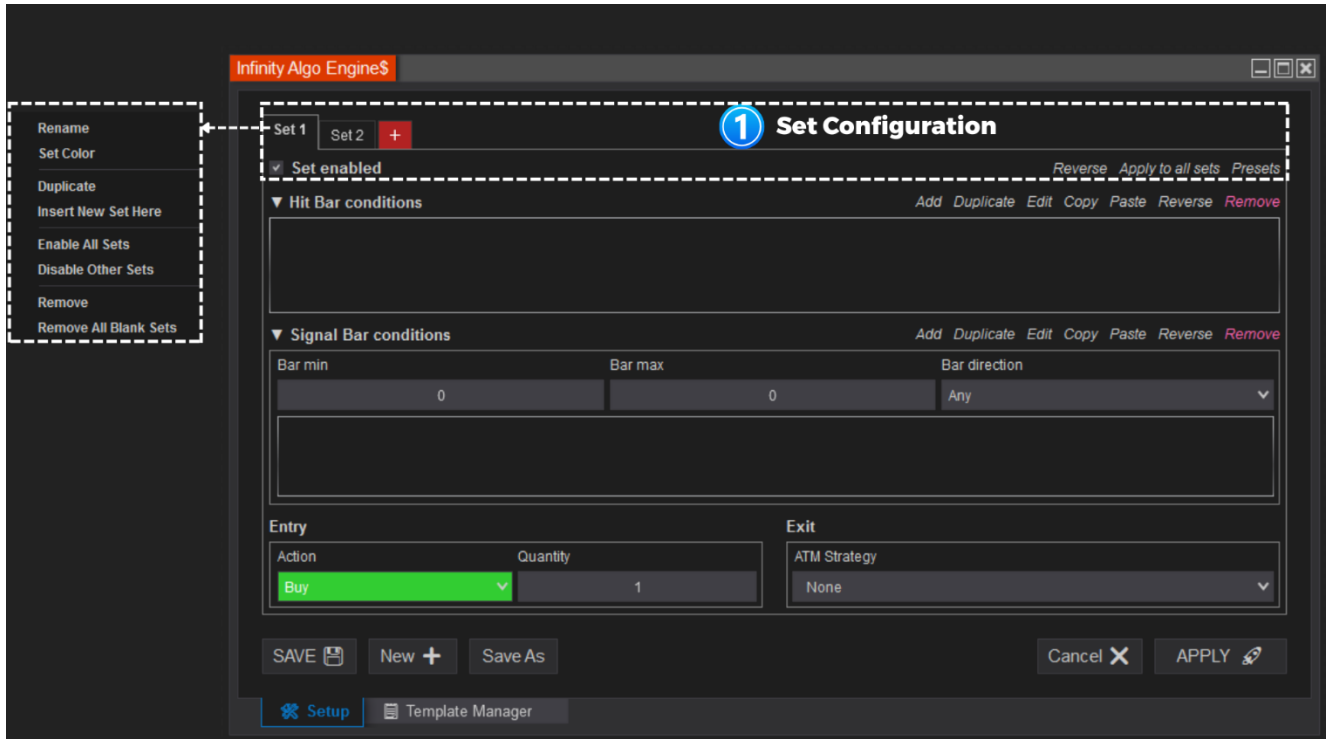
Additionally, a negative offset can also be applied. See Section 5 below for more details.

- **Note:**
 - When using "**MKT**" (**Market Order**), the system automatically ignores the Reference Price and Reference Price: Offset Value settings.
 - When using other order types (non-MKT), the system applies these settings to determine the placement of the pending order, based on the selected Reference Price and the Offset Value relative to the closing price of the signal bar.
- **Row 6:** Selects the trading account used by the system.

2.1.2 "Set up" Tab

The "**Set up**" **Tab** is where all signal conditions of the system are configured and managed. It defines **when the system is allowed to enter a trade**. Each Set represents a complete group of conditions, including:

- Hit Bar conditions
- Signal Bar conditions
- Entry
- Exit



- **Set Configuration**

This section is used to manage and configure the system's condition sets (**Sets**).

Each **Set** represents an independent group of conditions. Infinity Algo Engine\$ allows multiple Sets to operate simultaneously, with each Set evaluated independently according to its own configuration.

- **Set 1, Set 2, ...:** Select the Set to edit. The system supports multiple Sets running in parallel.
- **Add (+):** Creates a new Set.
- **Set enabled:** Enables or disables the current Set. When disabled, the Set will not be evaluated by the system.
- **Reverse:** Reverses the conditions and/or actions of the current Set.

- **Apply to all sets:** Applies the current Set's configuration (Quantity, Bar min, Bar max, etc.) to all other Sets.
- **Presets:** Saves or loads configurations for Signal Bar, Entry, and Exit.

When you **right-click on a Set**, the system displays the following quick management options:

- **Rename:** Renames the current Set.
- **Set Color:** Sets the display color for the Set.
- **Duplicate:** Creates a copy of the current Set with all its condition configurations.
- **Insert New Set Here:** Inserts a new Set at the position of the selected Set.
- **Enable All Sets:** Activates all Sets in the list.
- **Disable Other Sets:** Disables all other Sets, leaving only the current Set active.
- **Remove:** Deletes the current Set.
- **Remove All Blank Sets:** Deletes all Sets that have no configured conditions.

Infinity Algo Engine\$

Set 1 Set 2 +

☒ Set enabled Reverse Apply to all sets Presets

▼ Hit Bar conditions Add Duplicate Edit Copy Paste Reverse Remove

2 Hitbar conditions

▼ Signal Bar conditions Add Duplicate Edit Copy Paste Reverse Remove

Bar min	Bar max	Bar direction
0	0	Any

Entry

Action	Quantity
Buy	1

Exit

ATM Strategy
None

SAVE New Save As Cancel APPLY

Setup Template Manager

- **Hit Bar conditions**

This section contains the conditions used to identify the **Hit Bar**. Infinity Algo Engine\$ applies **AND logic**, meaning all conditions listed under Hit Bar Conditions must be satisfied.

The **Hit Bar** is the bar that defines the initial condition for the system to begin searching for a Signal Bar. If no Signal Bar conditions are configured and the default settings are kept, the system will treat the **Signal Bar** as the **Hit Bar**.

Available actions:

- **Add:** Adds a new condition.
- **Duplicate:** Creates a copy of the selected condition.
- **Edit:** Modifies the selected condition.
- **Copy / Paste:** Copies and pastes conditions.
- **Reverse:** Reverses the selected condition.
- **Remove:** Deletes the selected condition.

The screenshot displays the 'Infinity Algo Engine\$' window. At the top, there are tabs for 'Set 1' and 'Set 2', with a '+' button to add more sets. Below the tabs, a 'Set enabled' checkbox is checked. To the right of this are links for 'Reverse', 'Apply to all sets', and 'Presets'. The main configuration area is divided into two sections: 'Hit Bar conditions' and 'Signal Bar conditions'. The 'Hit Bar conditions' section is currently empty. The 'Signal Bar conditions' section is highlighted with a dashed white border and contains a blue circle with the number '3' next to the text 'Signal Bar conditions'. Below this, there are three input fields: 'Bar min' (set to 0), 'Bar max' (set to 0), and 'Bar direction' (set to 'Any'). At the bottom of the window, there are two sections: 'Entry' and 'Exit'. The 'Entry' section has an 'Action' dropdown set to 'Buy' and a 'Quantity' input set to '1'. The 'Exit' section has an 'ATM Strategy' dropdown set to 'None'. At the very bottom, there are buttons for 'SAVE', 'New', 'Save As', 'Cancel', and 'APPLY'. A 'Setup' button is also visible in the bottom left corner.

- **Signal Bar conditions**

This section contains the conditions used to identify the **Signal Bar**. Infinity Algo Engine\$ applies **AND logic**, meaning all conditions listed under **Signal Bar Conditions** must be satisfied.

The **Signal Bar** is the bar that appears after the **Hit Bar** and meets the defined **Signal Bar Conditions**.

- **Bar min:** The minimum number of bars from the **Hit Bar**.
- **Bar max:** The maximum number of bars from the **Hit Bar**.

The **Signal Bar** must fall within the range **Bar min → Bar max** (counted from the Hit Bar) and is evaluated only after a valid Hit Bar has been identified.

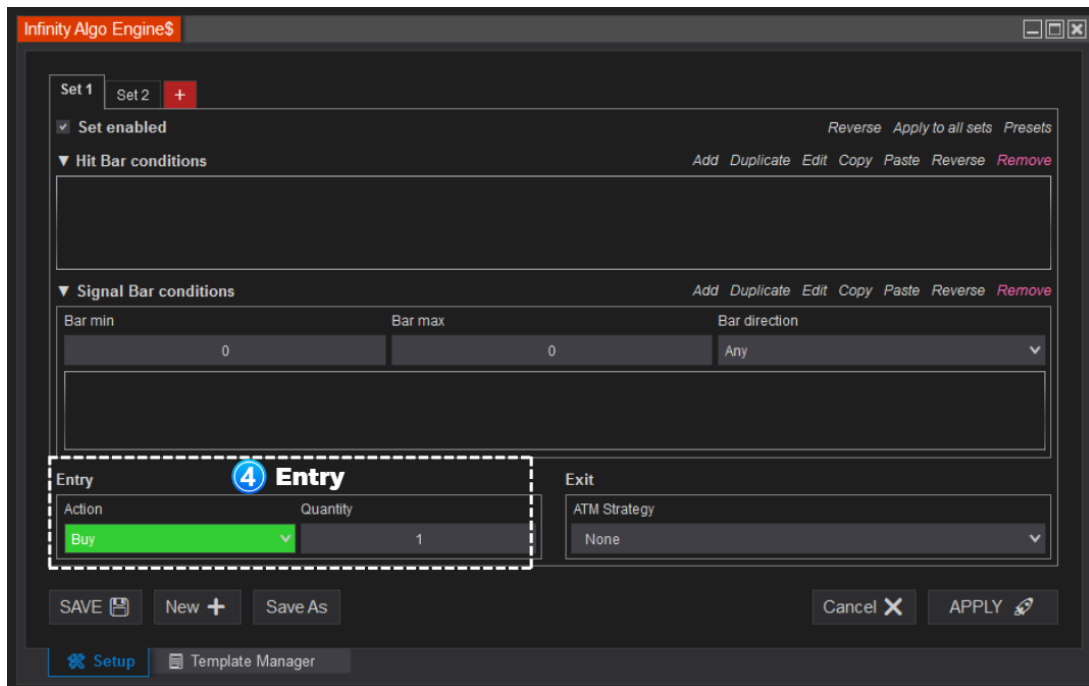
Example:

If **Bar Min = 3** and **Bar Max = 10**, this means:

The earliest bar that can be identified as the Signal Bar is the 3rd bar after the Hit Bar, and the latest possible Signal Bar is the 10th bar after the Hit Bar.

The range between these two bars defines the search window where Infinity Algo Engine\$ looks for a valid Signal Bar.

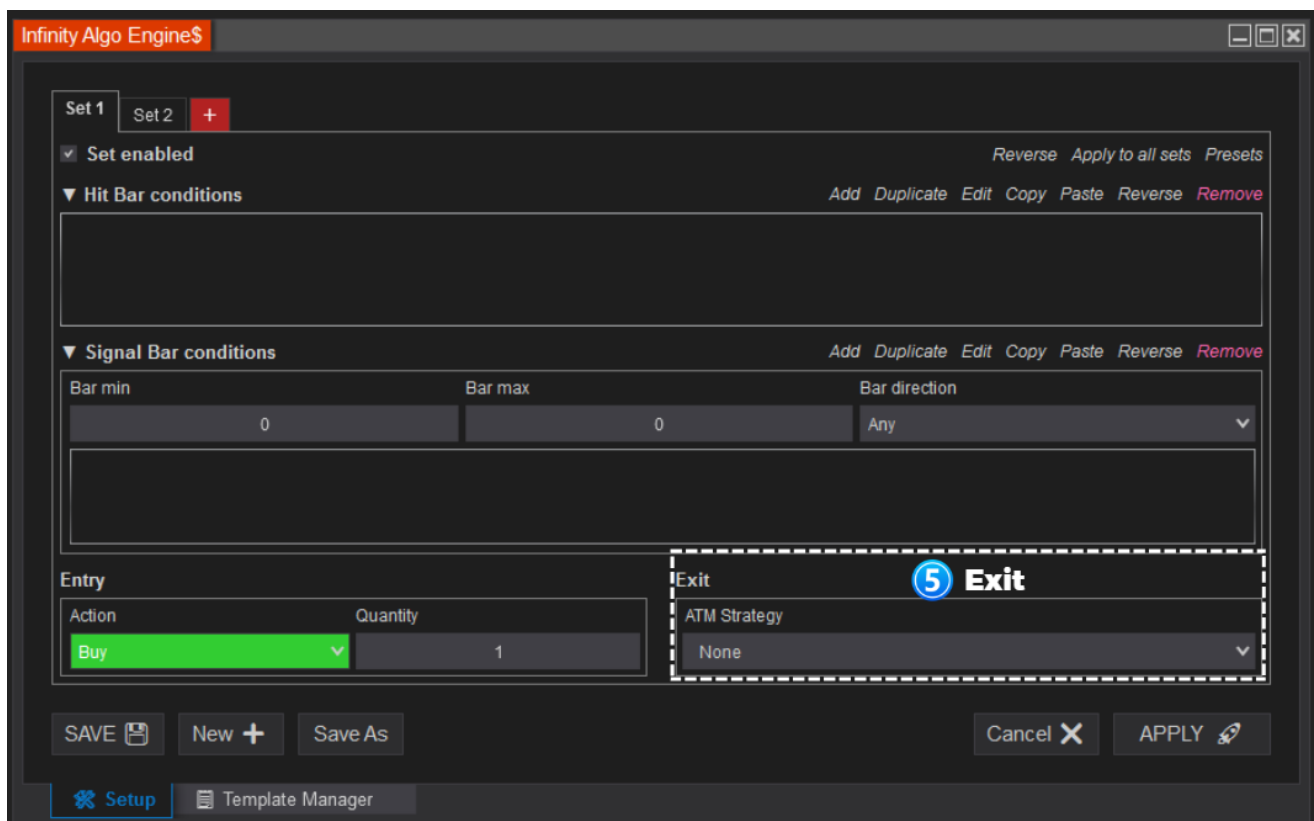
- **Bar direction:** Specifies the direction of the Signal Bar.
 - Any: Any bar after the Hit Bar that satisfies the Signal Bar conditions.
 - Up: A bullish bar after the Hit Bar that satisfies the Signal Bar conditions.
 - Down: A bearish bar after the Hit Bar that satisfies the Signal Bar conditions.



- **Entry**

- **Action:** Selects the action when the conditions are met (Buy / Sell).
- **Quantity:** The number of contracts.

The Entry is triggered after a valid Hit Bar is identified and the Signal Bar meets the defined conditions. If the LONG or SHORT button is enabled on the control panel, the system will activate the Entry and place the order automatically.

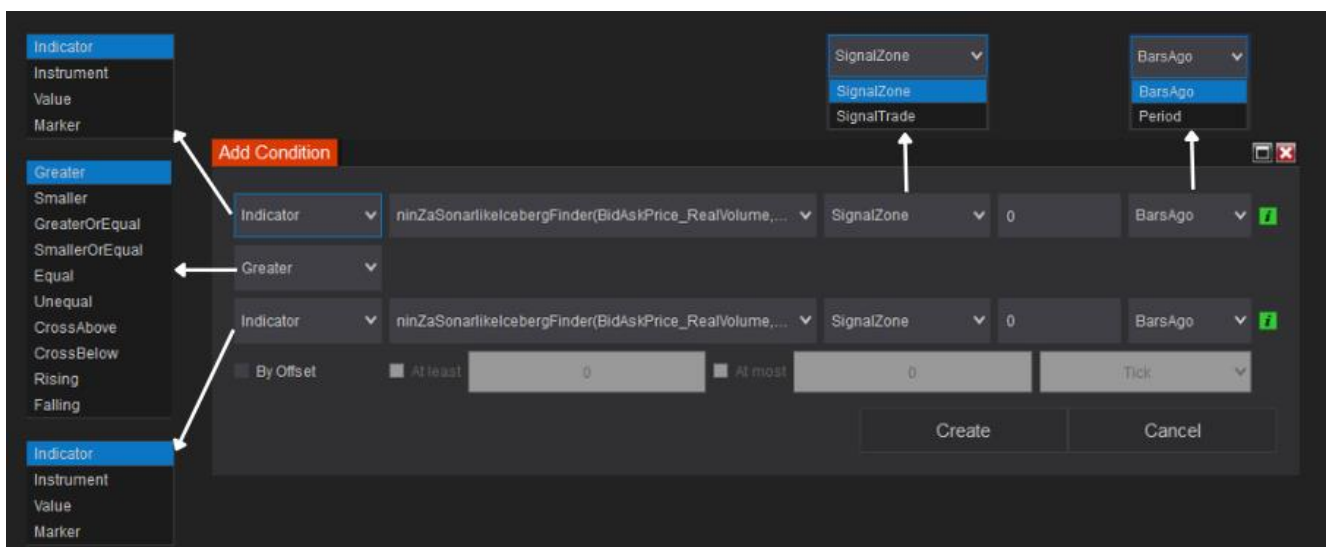
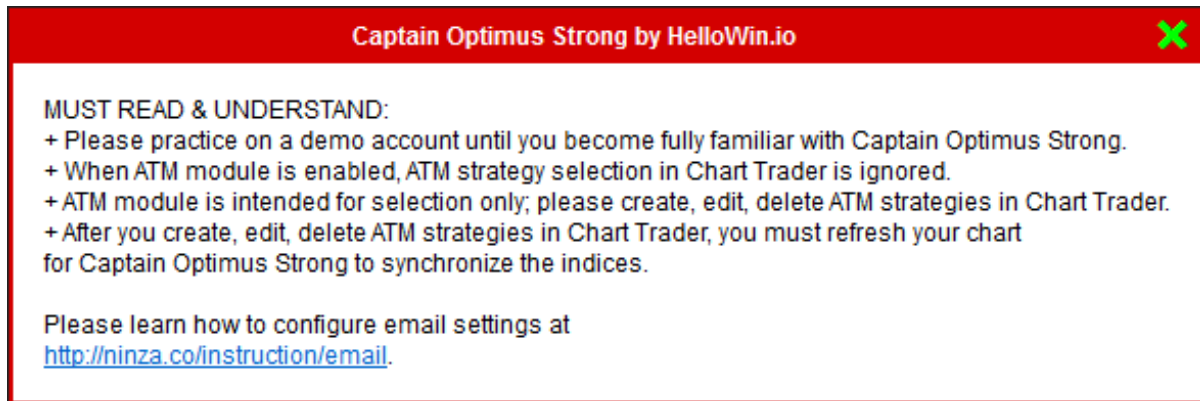


- **Exit**

- **Action: ATM Strategy:** Select the ATM strategy to be used.

If ATM = None, the system will not apply automated trade management after the order is executed.

- **Note:** When Infinity Algo Engine\$ is launched, a **note panel** will appear containing important information related to configuring your ATM strategy. Please review this information carefully.



- **Condition Builder**

When Add or Edit is selected in Hit Bar Conditions or Signal Bar Conditions, the Condition Builder window opens. The system supports four types of objects for building comparison conditions:

- **Indicator:** Data generated by an indicator on the chart, typically plots or signals used as inputs for condition configuration.

Common data examples:

- Signal / State (bullish, bearish, neutral, etc.)
- Plot value (e.g., moving average lines, oscillator values)

- Trailing Stop line
- Threshold line
- Any dedicated NinjaScript signal provided by the indicator

Notes:

- For our indicators, you can refer to the “**Dedicated NinjaScript Signals**” section on the product description page (ninZa.co, Renkokings.com) to select the correct data type for your conditions.
 - For 3rd-party indicators, you need full information about the **plots or signals** provided by that indicator in order to select the correct input for condition configuration.
- **Instrument:** Price data from the instrument currently being traded, used as inputs for condition configuration.

Common data types include:

- Open
 - High
 - Low
 - Close
 - Median / Typical (if supported by the system)
 - Volume (depending on configuration)
- **Value:** A fixed numeric value entered by the user, which can be an integer or a decimal.

Value is typically used to:

- Define comparison thresholds
 - Specify an indicator signal state
 - Compare with data from an Indicator or Instrument
- **Marker:** A reference point identified by indicators on the chart, used as an input for condition configuration.

Infinity Algo Engine\$ can detect all text-based markers generated by indicators running on the chart, including:

- Text characters
- Symbols
- Colors

Markers are typically used when an indicator displays signals directly on the chart but does not provide plot data or NinjaScript signals.

- **Comparison Operators:** These are the operators used to compare data between 2 objects when building conditions in the Condition Builder, including:
 - Greater
 - GreaterOrEqual
 - Equal
 - Smaller
 - SmallerOrEqual
 - Unequal
 - CrossAbove
 - CrossBelow
- **Bars-ago value:** Defines the position of a bar, or a group of **N bars**, used to retrieve data for comparison.

The system supports 4 comparison modes based on bar positions:

- **Bar Ago vs. Bar Ago:** A specific bar is compared with another specific bar.
- **Bar Ago vs. Period:** A specific bar is compared with each bar within a range of N bars.
- **Period vs. Bar Ago:** Each bar within a range of N bars is compared with a specific bar.
- **Period vs. Period:** 2 ranges of N bars are compared in parallel, with each bar matched to the corresponding bar in the other range.
- **By Offset:** Allows you to define a difference (offset) between two values being compared. Instead of comparing two objects directly, the system compares them using a specified offset distance.

Available options:

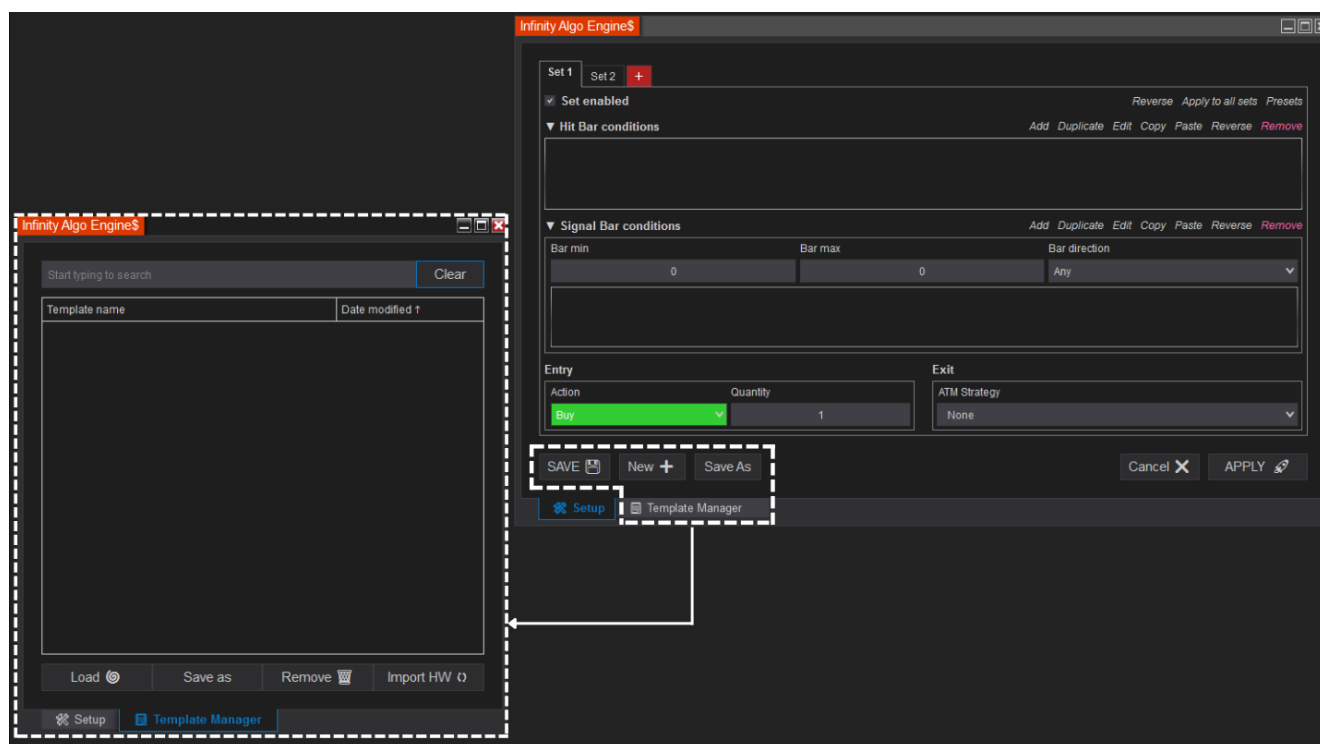
- **At least:** The minimum difference between the 2 values.
- **At most:** The maximum difference between the 2 values.

The Offset unit can be configured as:

- **Tick:** The difference measured in ticks.
- **Point:** The difference measured in points.

Offset is typically used when a value must be **greater or smaller than another value by a specific amount**, rather than simply being greater than or equal to it.

2.1.3 Template Manager Tab



The Template Manager allows you to store and manage condition configurations created in the “Set up” Tab.

Each Template includes the complete configuration of:

- Condition Sets
- Hit Bar conditions
- Signal Bar conditions
- Entry
- Exit

Functions available in Template Manager:

- **Search:** Searches for templates by name.
- **Load:** Loads the selected template into the system.
- **Save As:** Saves the current configuration as a new template.
- **Remove:** Deletes the selected template.
- **Import:** Imports templates from the Hellowin Backtest Tool (if available).

Template list displays:

- Template name
- Last modified date

2.2 Core Modules

2.2.1 Execution Core

- **Input Setup:** Before the system can identify Hit Bars and Signal Bars, you need to configure the input data used for the conditions.

Standard workflow:

- **Step 1:** Add indicators to the chart (the indicators you want to use as conditions).
- **Step 2:** Open the Set up Tab and create a Set (Set 1, Set 2, etc.).
- **Step 3:** Add conditions to Hit Bar Conditions and/or Signal Bar Conditions.

In the **Condition Builder**, select:

- Indicator / Instrument / Value
- The plot or data to compare
- Bars-ago
- Operator (>, <, =, CrossAbove, CrossBelow, etc.)
- **Step 4:** Configure Bar min / Bar max / Bar direction if Signal Bar is used.
- **Step 5:** Configure Entry / Exit (Action, Quantity, ATM, etc.).

After completing the Input Setup process and selecting Apply, the system will calculate and display the Hit Bar and Signal Bar on the chart when all conditions are satisfied.

- **Auto Trading**

After completing the Input Setup, follow the steps below to configure Infinity Algo Engine\$ for automated trading:

- **Step 1:** Select the trading account for automated execution in Row 6.
- **Step 2:** Select the order type you want to use to enter the market in Row 5.
 - If MKT (Market Order) is selected, you do not need to adjust the Reference Price (Row 4) or Reference Price: Offset Value (Row 5).
 - For other order types, you can adjust Reference Price (Row 4) and Reference Price: Offset Value (Row 5) according to the price level where you want Infinity Algo Engine\$ to execute the trade.
- **Step 3:** Review all entry and exit conditions in Input Setup, verify historical Hit Bar and Signal Bar signals on the chart, and check the risk management settings in the Money Management section to ensure the system behaves as intended.
- **Step 4:** In Row 3, enable the "LONG" and/or "SHORT" buttons. When a signal is generated, the system will automatically execute the corresponding long or short order and begin the automated execution process.

To stop automated trading, disable the "LONG" and/or "SHORT" buttons.

- **Execution Flow Model**

After completing the Input Setup and enabling Auto Trading, Infinity Algo Engine\$ processes signals and executes trades according to the following workflow:

Indicator (provides input data for the system)



Hit Bar (checks conditions to begin signal detection)



Signal Bar (confirms the trading signal)



Entry Proposal (displays the proposed entry location)



Order Routing (calculates the price and submits the order based on the selected Order Type)



Trade Behavior (applies trading state rules)



ATM (manages Stop-Loss / Take-Profit if used)



Money Management (controls Daily Profit / Daily Loss limits)

2.2.2 Order Routing & Offset

This module is responsible for calculating the order price and submitting orders to the market according to the configured settings.

The process consists of 2 main steps:

1. Determine the order price based on the **Reference Price** and **Offset**.
2. Submit the order to the market based on the selected **Order Type**.

- **Offset Computation**

Offset is an adjustment value (measured in ticks or points) that is added to or subtracted from the **Reference Price** to determine the order placement. This allows users to control the distance between the signal and the actual entry price.

General formula: **Order Price = Reference Price ± Offset**

Adjusting the Offset helps adapt the entry placement to different market conditions such as pullbacks, breakouts, or high volatility environments.

- **Order Routing**

After the order price is calculated, the system submits the order to the market according to the selected **Order Type**.

Infinity Algo Engine\$ supports the following order types:

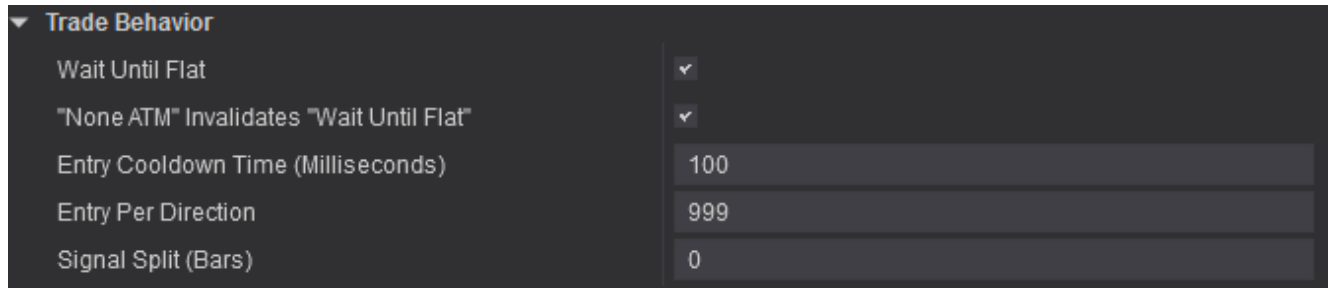
- **MKT (Market Order)**: Executes immediately at the current market price.
- **LMT (Limit Order)**: Executes at the specified price or better.
- **MIT (Market If Touched)**: When the price touches the specified level, a Market Order is sent.
- **STP (Stop Order)**: Triggers when the price breaks through a specified level.
- **SLM (Stop Limit Order)**: When triggered, the system places a Limit Order instead of a Market Order.

Each order type has a different trigger mechanism and is suited for different execution strategies.

2.2.3 Trade Behavior

Trade Behavior manages the trading state and defines the conditions under which new trades are allowed.

It includes the following mechanisms:



Trade Behavior	
Wait Until Flat	☒
"None ATM" Invalidates "Wait Until Flat"	☒
Entry Cooldown Time (Milliseconds)	100
Entry Per Direction	999
Signal Split (Bars)	0

- **Wait Until Flat**

When this feature is enabled, the system is allowed to enter a new trade only when **there are no open positions**. This helps prevent overlapping positions and ensures that each signal generates only one trade.

- **"None ATM" Invalidates "Wait Until Flat"**

When this feature is enabled and the ATM Strategy is set to None, the Wait Until Flat rule is no longer enforced. When a new signal appears, the system will **automatically close the current trade** and open a new trade according to the configured settings.

- **Entry Cooldown Time (Milliseconds)**

This parameter defines the minimum time interval between two consecutive entries, measured in milliseconds.

It helps prevent the system from triggering multiple trades within a very short period when the market is highly volatile or when signals appear close together.

- **Signal Split (Bars)**

This parameter specifies the minimum number of bars required between two signals for the signal to be considered valid.

If a new signal appears within fewer bars than this value, the system will ignore that signal.

- **Entry Per Direction**

This parameter defines the maximum number of consecutive entries per direction (Buy or Sell).

It helps control how many trades can be executed in the same direction during a sequence of signals.

Objectives of the Trade Behavior module:

- Control trading frequency.
- Prevent consecutive entries caused by signal noise.
- Manage the position state before executing a new order.

2.2.4 ATM Integration Layer

This module integrates directly with **NinjaTrader's ATM Strategy** to automatically manage **Stop-Loss** and **Take-Profit** according to the configured settings.

The **ATM configuration** is located in **Set up Tab → Exit**. Since each Set represents a complete trading configuration, each Set can use its own ATM Strategy.

This allows you to:

- Apply the same ATM strategy across all Sets, or
- Use different ATM strategies for different trading setups.

There are 2 main modes:

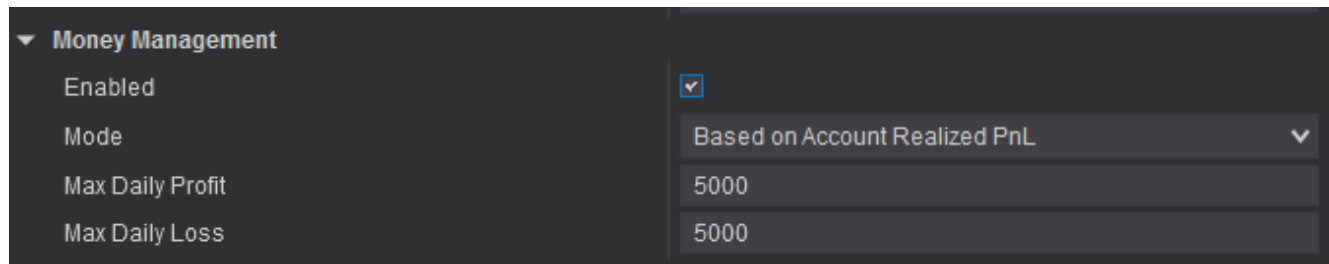
- **ATM = None:** The system does not use the ATM order management mechanism.
 - If "**None ATM**" Invalidates "**Wait Until Flat**" is enabled, the system will close the current open position when a new signal appears and open a new trade according to the configured settings.

This is also the default configuration of the system.
 - If "**None ATM**" Invalidates "**Wait Until Flat**" is disabled, open positions must be closed manually, and the system will open a new trade only after the previous position has been closed.
- **ATM = Custom:** The user utilizes the ATM Strategy for order management. In this case:
 - Stop-Loss and Take-Profit for the position are automatically set according to the ATM configuration.
 - The ATM strategy manages the entire lifecycle of the trade until the position is closed.

- If **Wait Until Flat is enabled**, the system will **open a new trade** only after the previous trade has been completed according to the ATM rules.
- If **Wait Until Flat is disabled**, the system will **close the current trade** when a new signal appears and open a new trade according to the configured settings.

2.2.5 Money Management

Money Management is the module that monitors daily profit and loss.

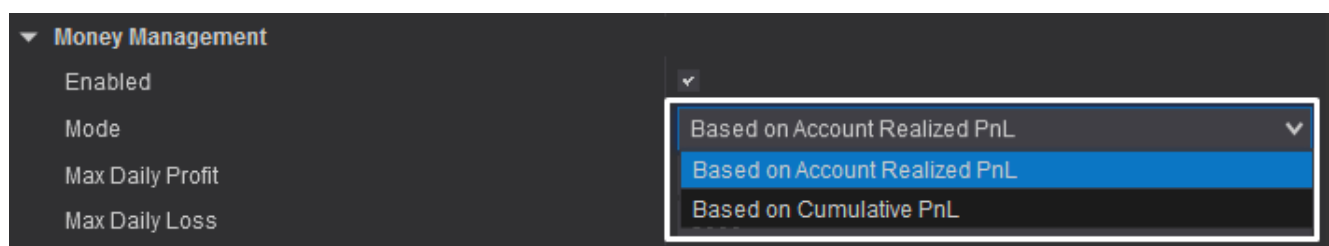


Money Management	
Enabled	☒
Mode	Based on Account Realized PnL
Max Daily Profit	5000
Max Daily Loss	5000

When the Daily Profit or Daily Loss threshold is reached:

- The system closes all open positions.
- The system stops executing new orders.
- Once activated, the system will automatically monitor and enforce the daily profit/loss limits based on the rules you configure.

2 modes are available within this feature:



Money Management	
Enabled	☒
Mode	Based on Account Realized PnL Based on Account Realized PnL Based on Cumulative PnL
Max Daily Profit	
Max Daily Loss	

- "Based on Account Realized PnL": If this mode is enabled, the Infinity Algo Engine\$ will flatten the current account once the account's Realized PnL reaches the Daily Profit or Daily Loss threshold.
- "Based on Cumulative PnL": If this mode is enabled, Infinity Algo Engine\$ will flatten all current positions on the account when the Cumulative PnL of Infinity Algo Engine\$ alone reaches the established Daily Profit or Daily Loss threshold.

Note:

When you select the "Based on Cumulative PnL" option, the system will also provide the "End Trading Day Reset Cum. PnL" feature.

When enabled, this feature resets the daily profit/loss tracking back to zero at the start of a new trading day, allowing the system to apply the daily profit and loss limits again for the new session.

Money Management	
Enabled	☒
Mode	Based on Cumulative PnL
Max Daily Profit	5000
Max Daily Loss	5000
End Trading Day Reset Cum. PnL	☒

3. Remaining Parameters in General

General	
Price Step: Unit	Tick
Order: Type	MKT
Reference Price: Buy	Close_SignalBar
Reference Price: Sell	Close_SignalBar
Reference Price: Offset	2
Marker: Auto Split Enabled	☒
Marker: Auto Split Splitter	#
Expand/Collapse Signal Bar Condition List Only	☒
Screen DPI	100

3.1 Price Step: Unit

- **"Price Step: Unit"**: This parameter defines the price step unit used in the system's calculations.

2 options are available:

- **Tick**: Uses the tick size of the instrument. This is the most common setting for futures markets.
- **ForexPip**: Uses pip units, typically applied in the Forex market.

This parameter affects how the system calculates **Offset values, price distances, and other conditions related to price increments**.

3.2 Marker

This group of parameters controls how the system detects and processes text-based markers on the chart when configuring **Input Setup**.

- **"Marker: Auto Split Enabled"**: When this option is enabled, the system automatically splits markers containing multiple pieces of information into separate elements based on a defined separator character.

This allows the **Condition Builder** to accurately identify each marker when configuring conditions.

- **"Marker: Auto Split Splitter"**: Defines the separator character used for automatic marker splitting.

Example:

If **Marker: Auto Split Splitter = #**, and the marker text is **"Buy#1"**, the system will split the marker into:

- BUY
- 1

3.2 Expand/Collapse Signal Bar Condition List Only

- **"Expand/Collapse Signal Bar Condition List Only"**: This option controls how the Signal Bar Conditions list is displayed in the Set up window.

When this option is enabled, the **Expand/Collapse** action applies only to the **Signal Bar Conditions list**, while the other sections of the Set remain unchanged.

This option allows users to focus on editing **Signal Bar conditions** without affecting the display layout of the other sections within the Set.